

DOCUMENT 07

Complete Validation & Experimental Program

L1–L5 Test Protocols · CFD Plans · SIL/HIL Framework · Statistics · Report
Templates

End-to-end validation programme for the UAV fleet ecosystem. Defines five validation levels from component bench to full system flight, complete SIL/HIL simulation environments using ROS2/Gazebo/PX4-SITL, rigorous statistical analysis framework with minimum sample sizes for every published claim, CFD validation methodology, and standardised test report templates.

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1. VALIDATION PHILOSOPHY

The validation programme follows the V-model systems engineering lifecycle. Every requirement in the system requirements document (DOC1 Sections III–IV) has a corresponding test that verifies it. The programme is sequenced left-to-right: decompose requirements to components, then integrate test evidence upward from components to system level. No system-level claim is published until it is traceable to at minimum one of: (a) bench measurement with calibrated instruments, (b) validated simulation that has itself been calibrated against bench, or (c) explicit qualification as a "model prediction — not yet validated" in the document.

Current validation status by contribution:

Contribution	Claimed Value	Validation Level	n	Gap
W ^{1.5} scaling law exponent	<0.1% error at 5 payload configs	L1 bench measurement, n=10 each	50 total	None — complete
FOC noise reduction at 5k RPM	13 dB SPL at 1m (anechoic)	L1 bench, anechoic chamber	10 per RPM	None — complete
FOC vibration reduction	11.3× at 5k RPM	L1 bench, strain gauges	10 per RPM	None — complete
CFRP safety factor 4.83	Under 3g load, FEA	L2 simulation (ANSYS FEA)	1 model	Physical mechanical test TBD
SOC estimation 0.9% RMS	CFRP vs AI EKF comparison	L2 simulation + bench	Limited	Outdoor dynamic flight not tested
SPF navigation 92.7%	Indoor course 15 trials	L5 indoor flight, 15 real trials	15	Minimum 97 needed for 95% CI
Docking success 99%	Lab, ≤5m/s wind	L5 lab, 100 trials	100	Outdoor ≥8m/s not yet tested (196 needed)
Fleet endurance 7+ hours	Scheduling model	Mathematical model only	—	Full hardware demonstration not done
IR beacon 500Hz guidance	Analytical derivation	L0 — analysis only	0	Complete L1–L5 validation required
Federated EKF 0.20m at 60s	Analytical derivation	L0 — analysis only	0	SIL simulation then flight required
CBBA-SOC fleet rotation	Algorithm design	L0 — design only	0	SIL simulation then flight required

Table 1. Validation status matrix — every contribution mapped to evidence level.

2. L1 — COMPONENT BENCH TESTS

2.1 Motor and ESC Characterisation Protocol

All motor/ESC characterisation uses a custom thrust stand: aluminium arm mount holding motor at one end, digital force gauge (resolution 1g) at the other, power analyser (accuracy 0.1% current, 0.1% voltage) logging at 1kHz, optical tachometer (±1 RPM), and calibrated K-type thermocouple on motor can.

Test ID	Test Name	Protocol	Instruments	Pass Criterion	n
M-01	Thrust curve (FOC)	Sweep throttle 10→100% in 5% steps, 10s hold each; log thrust, current, RPM, efficiency	Thrust stand + power analyser + tachometer	Max thrust ≥ 2.8 kg at $<10A$; hover efficiency $\geq 80\%$	5 per config
M-02	FOC vs trap noise	Record SPL at 1m in anechoic at 1k, 3k, 5k, 10k RPM both modes	GRAS 46AE mic + NI DAQ 96kHz; Hann FFT 65536pt	13 dB FOC vs trap at 5k RPM ± 1 dB	10 per RPM per mode
M-03	Vibration signature	Mount strain gauges on arm; record RMS at all RPMs, both modes	HBM LY13-3/120 + 1kHz data logger	11× vibration reduction FOC vs trap at 5k RPM	10 per RPM per mode
M-04	ESC thermal saturation	Run at hover PWM in 25°C and 40°C ambient 60min continuous	FLIR ONE thermal camera; thermocouple on FET	FET $< 80^\circ\text{C}$ at 25°C; $< 95^\circ\text{C}$ at 40°C with fan	3 per temp
M-05	Motor efficiency map	2D sweep: voltage (3S, 4S, 5S) $\times$ throttle (20–100%); log $\eta = P_{\text{mech}}/P_{\text{elec}}$	Power analyser + torque meter (calibrated spring)	$\geq 85\%$ peak efficiency; identify optimal operating point	3 per voltage
M-06	Current THD measurement	FFT of motor current waveform at hover RPM, FOC vs trap	Rogowski current probe + 96kHz DAQ	THD_FOC $< 3\%$; THD_trap $> 15\%$	5 per mode

Table 2. Motor/ESC L1 test protocols.

2.2 Battery Characterisation Protocol

Test ID	Test Name	Protocol	Pass Criterion	n
B-01	Rated capacity at 1C	Full CC-CV charge; 1C constant current discharge to 3.0V/cell; log Wh	$\geq 95\%$ rated capacity (148 Wh for W1 Li-ion)	5 new cells; 3 at 50 cycles; 3 at 150 cycles
B-02	Capacity at 5C	Same as B-01 at 5C discharge rate	$\geq 88\%$ rated capacity (Peukert effect)	5 cells
B-03	Internal resistance vs SOC	EIS at 0.1Hz–100kHz at 10/30/50/70/90% SOC	$R_{\text{int}} < 15 \text{ m}\Omega/\text{cell}$ at 50% SOC; $< 25 \text{ m}\Omega$ at 20%	3 cells
B-04	Temperature coefficient	B-01 repeated at 5°C, 10°C, 25°C, 45°C ambient in temp chamber	Capacity at 10°C $\geq 85\%$ of 25°C baseline	2 cells per temp
B-05	Cycle life degradation	200 cycles at 2C; measure capacity every 20 cycles	$\geq 80\%$ initial capacity at cycle 200	2 packs minimum
B-06	Thermal runaway threshold (LiFePO4)	Nail penetration test on single cell (no drone)	No fire; max temp $< 200^\circ\text{C}$; must confirm LiFePO4 safety	1 cell (destructive)
B-07	Voltage sag under combined load	Propulsion (3.67A) + compute (1.67A) simultaneous at 50% SOC	Sag $< 0.9V$ (confirmed isolated rail eliminates brownout)	10 events

Table 3. Battery characterisation test protocols.

2.3 IR Beacon Guidance System — Component Tests

Test ID	Test Name	Protocol	Pass Criterion
IR-01	Lateral sensitivity map	PI XY stage moves IRDG-01 board vs DOCB-01; measure ϵ_x at 5mm increments across $\pm 80\text{mm}$ at $h=100,200,300\text{mm}$	$ d\epsilon_x/dx \geq 0.3 \text{ mm}^{-1}$ at $h=200\text{mm}$; nonlinearity < 5% over $\pm 50\text{mm}$
IR-02	Update rate verification	Oscilloscope on IRDG-01 UART output TX pin; measure period between packets	$\geq 500 \text{ Hz}$ nominal; jitter < 2ms; no dropouts in 60s
IR-03	Frequency isolation (crosstalk)	Block LED1 physically; verify ϵ_x changes as expected; repeat for each LED	Cross-channel leakage < 1% between any pair
IR-04	Sunlight immunity	Repeat IR-01 under 1000 W/m^2 solar simulator (Oriel 91191-1000)	Sensitivity degradation < 15%; no signal loss
IR-05	Rain immunity	Mist nozzle over DOCB-01 board during IR-01; maintain water on lens surface	No signal loss; alignment error < 3mm
IR-06	Temperature drift	Repeat IR-01 at 5°C and 50°C ambient; record sensitivity change	Sensitivity variation < 10% over $5\text{--}50^\circ\text{C}$ range
IR-07	Bearing/heading sensitivity	Rotate drone $\pm 30^\circ$ yaw above dock; verify ϵ_x/ϵ_y still provide valid guidance	Valid guidance within $\pm 25^\circ$ yaw misalignment

Table 4. IR docking guidance system L1/L2 test protocols.

3. L3 — SOFTWARE-IN-THE-LOOP (SIL) ENVIRONMENT

3.1 SIL Stack Specification

Layer	Software	Version	Purpose
OS	Ubuntu 22.04 LTS	22.04.3	Host OS for all simulation
Middleware	ROS 2 Humble	2.0 LTS (Humble Hawksbill)	Inter-process communication, topic/service infrastructure
Flight stack	PX4 SITL	v1.14.3	Flight controller simulation on PC; DSHOT/PWM virtual outputs
Simulator	Gazebo Classic 11	11.10	3D physics simulation; wind models; IR beacon plugin
Ground control	QGroundControl	v4.3.0	Mission planning, parameter setting, telemetry monitoring
Custom plugins	UAV Fleet Sim package (custom)	v0.1 (to be developed)	IR beacon simulation; UWB ranging noise model; fleet scheduling node
Battery model	Custom Python Thevenin EKF model	—	Accurate SOC simulation with voltage sag under combined load
Navigation AI	SPF framework (ROS2 port)	Research code	Vision-language navigation testing

Table 5. SIL software stack specification.

3.2 SIL Custom Plugin Requirements

The following Gazebo plugins must be developed before SIL testing can proceed:

- **IR Beacon Plugin:** Gazebo world plugin that simulates four IR LED sources at dock position; computes differential intensity at simulated photodetector positions on drone underside; publishes $[\epsilon_x, \epsilon_y, \text{height}]$ on ROS2 topic matching real IRDG-01 UART output format.
- **UWB Ranging Plugin:** Simulates Decawave DWM3000 two-way ranging between drone pairs; adds configurable Gaussian noise ($\sigma=0.1\text{m}$ default) and multipath bias; publishes range messages consumed by federated EKF node.
- **Federated EKF Node:** ROS2 node implementing information-form EKF fusion of UWB ranges; publishes worker position estimates; measures divergence vs ground truth from Gazebo.
- **CBBA-SOC Node:** ROS2 node implementing the auction algorithm with SOC weight function; subscribes to /battery_state for each drone; publishes task assignments; logs convergence rounds.
- **Battery Sim Node:** Accurate LiPo/Li-ion simulation using Thevenin model with $R_{\text{int}}(\text{SOC}, T)$; consumes motor power from ESC commands and compute load; publishes SOC and voltage.

3.3 SIL Test Matrix

Test ID	Scenario	Drones	Duration	Metric	Pass Criterion	Runs
S-01	Single drone autonomous waypoint circuit	1 W2	5min	Waypoint success %, position error, battery at RTH	$\geq 98\%$ success; RTH SOC $\geq 35\%$	50
S-02	CBBA-SOC rotation, 5 workers 2 docks	5 W1 + 1 C1 + 1 K1	60min	Battery failures, task completion, cycle time	0 battery failures; cycle $\leq 35\text{min}$	20
S-03	IR-guided docking at 0/4/8/12 m/s	1 W1 + 1 D1 (simulated wind)	per trial	Success rate, alignment error distribution	$\geq 95\%$ at 8m/s; $\geq 80\%$ at 12m/s	50 per wind speed
S-04	Federated EKF GPS-denied	5 W1, GPS off at $t=2\text{min}$	10min GP S-denied	σ_{pos} vs time; drift rate	$\sigma_{\text{pos}} \leq 0.3\text{m}$ at $t=5\text{min}$	10
S-05	Command node failure mid-mission	5 W1 + 1 C1; kill C1 at $t=5\text{min}$	15min	Failover latency; mission completion post-failover	Failover $< 15\text{s}$; $\geq 70\%$ mission complete	10
S-06	ESP-NOW node scaling stress test	5/10/15/20 nodes	30min	Mesh latency, packet loss, TDMA performance	Latency $< 50\text{ms}$ at ≤ 15 nodes; document 20-node behaviour	5 per fleet size
S-07	Multi-carrier relay coordination	5 W1 + 2 K2 + 1 K1 (Phase 2)	120min	K2 SOC management, handoff latency	Zero battery failures; K2 handoff $< 90\text{s}$	5

Table 6. SIL test matrix with pass criteria.

4. L4 — HARDWARE-IN-THE-LOOP (HIL)

HIL connects real flight controller hardware to the PX4 SITL environment via serial link. The Pixhawk 6C runs actual production firmware; SITL provides simulated sensor data (IMU, GPS, barometer, magnetometer) over a USB-serial bridge at 921600 baud. ESC commands output from the FC are captured and fed to the Gazebo motor model. This allows PID gain validation, failsafe logic verification, and power management testing on real hardware before any propellers are attached.

Test ID	Test Name	Setup	Protocol	Pass Criterion
H-01	PID baseline tuning	Pixhawk 6C + SITL on W1 mass model	Step response test: 30° roll, 30° pitch, 180° yaw commands; measure overshoot, settle time	Overshoot < 20%; settle < 0.8s; phase margin > 45°
H-02	Failsafe logic verification	Pixhawk + SITL	Inject 6 failsafe conditions (RTH, land, hold, etc.) via MAVLink parameter injection	All failsafes trigger correct action within 500ms
H-03	Power architecture isolation test	Real PM02D + real battery + real Pixhawk on bench	Inject 5.34A combined load spike; measure FC voltage at PM02D output	FC voltage never drops below 4.8V; 0 brownouts in 100 spike events
H-04	SOC estimation calibration	Real battery + BMS + Pixhawk power module	Discharge at known rate; compare EKF SOC vs true SOC (measured Ah)	SOC error < 2% RMS over full discharge cycle
H-05	Motor failure response	Pixhawk + SITL; inject motor failure at t=10s	Verify 3-motor mode activates; measure attitude deviation from commanded	Attitude deviation < 15° within 500ms of failure; safe descent initiated
H-06	GPS spoofing detection	Pixhawk + SITL; inject inconsistent GPS velocity	Inject GPS position jump 50m at t=10s with smooth IMU	SPOOF flag within 2s; GPS ignored; switch to cooperative loc state

Table 7. HIL test protocols.

5. L5 — SYSTEM FLIGHT TEST CAMPAIGN

5.1 Site Requirements

- Open field minimum 200×200m; no overhead obstacles within 150m radius of launch point
- DGCA-compliant airspace (Class G uncontrolled below 400ft; outside 5km of any airport)
- Vicon motion capture (8-camera) installed in 10×10×5m indoor volume for precision measurement
- Wind measurement: Davis Vantage Pro ultrasonic anemometer at drone operating altitude (≥10m mast)
- Emergency response kit on site: fire extinguisher (CO2 class); first aid kit; drone retrieval poles
- Minimum flight crew: 1 pilot in command (DGCA RPC holder), 1 safety observer, 1 GCS operator

5.2 Flight Test Phase Programme

Phase	Conditions	Tests	Min Trials	Go Criterion for Next Phase
F1 — Indoor tethered	Vicon arena; 3m tether; <1m/s indoor air	Hover stability, attitude response, vibration logging, power measurements	10 flights	Position error <10cm; vibration <0.5mm/s; zero brownouts
F2 — Outdoor hover basic	Open field; 0–4m/s wind; 5m AGL; VLOS	GPS hold, manual override latency, RTH accuracy, FOC noise measurement	20 flights	Hold ±1m; RTH within 3s; FOC noise matches bench data ±2dB
F3 — Autonomous waypoints	Open field; 0–6m/s; 30m AGL; 500m circuit	Full autonomous mission; energy model validation; SOC accuracy log	15 flights	Waypoint ±3m; energy model error <5%; all failsafes verified
F4 — Docking trials (calm)	K1 deployed; 0m/s; calm day	50 autonomous landing attempts on K1	50 trials	≥95% success; ≤20mm mean alignment error; cycle ≤35min

Phase	Conditions	Tests	Min Trials	Go Criterion for Next Phase
F5 — Docking trials (wind)	K1 deployed; 4m/s and 8m/s wind (anemometer confirmed)	30 autonomous landings per wind condition	60 trials total	≥92% at 4m/s; ≥85% at 8m/s; characterise failure modes
F6 — 2-drone fleet rotation	2 W1 + K1 on mains; 500m survey area	CBBA-SOC rotation; battery fault injection; complete mission coverage	20 rotation cycles	Zero battery failures; coverage gap ≤2min; cycle ≤35min
F7 — Full fleet endurance	5 W1 + C1 + K1; 2-hour continuous mission; all instruments	All mission metrics simultaneously; end-to-end demonstration	3 two-hour runs	Coverage ≥95% of time; no LOA events; all metrics in spec

Table 8. Flight test campaign phases with go/no-go criteria.

6. STATISTICAL ANALYSIS FRAMEWORK

6.1 Minimum Sample Size Requirements

Every quantitative performance claim requires a minimum sample size calculated from the desired confidence level and margin of error. The formula for a proportion estimate:

$$n \geq (z_{\{\alpha/2\}})^2 \times p(1-p) / \varepsilon^2$$

where $z_{\{\alpha/2\}} = 1.960$ for 95% confidence interval, p = claimed proportion, ε = acceptable margin of error.

Claim	p	ε	z	Required n	Current n	Status	Action
Docking success ≥99% (calm)	0.99	0.03	1.96	43	100	✓ Done	None needed
Docking success ≥95% (8m/s wind)	0.95	0.05	1.96	73	0	✗ MISSING	Must complete before outdoor claim
Docking success ≥85% (12m/s)	0.85	0.06	1.96	136	0	✗ MISSING	Phase F5 required
SPF navigation ≥92.7% real-world	0.927	0.05	1.96	97	15	✗ Insufficient	82 more trials needed
Brownout rate = 0% (with PM02D)	<0.01	0.05	1.96	~15	~30	✓ Done	Log maintained
W ^{1.5} scaling <0.1% error	—	<0.001	—	5 configs × 10	50	✓ Done	None
FOC 13dB noise reduction	—	±1 dB	—	10 per RPM	10 per RPM	✓ Done	None
CBBA-SOC <1% battery failure	<0.01	0.05	1.96	~15 runs	0	✗ Missing	SIL S-02 first; flight F6 after
Fleet endurance 7+ hours	—	—	—	3 full runs	0	✗ Missing	F7 required; needs carrier

Table 9. Statistical adequacy analysis for all major claims. MISSING items must be resolved before publication.

6.2 ANOVA and Post-Hoc Analysis Plan

For the hover power endurance table (5 payload configurations × 10 trials), one-way ANOVA with Tukey HSD post-hoc correction ($\alpha = 0.05$) confirms that each payload level is statistically distinguishable from adjacent levels. All

five pairwise p-values must be < 0.05 after Tukey correction. Current status: complete for the 5-configuration dataset ($n=50$ total). Same analysis repeated for FOC vs trapezoidal noise at each RPM level using unpaired two-sample t-test ($\alpha = 0.05$, two-tailed). Required: $p < 0.001$ for 13 dB difference — easily achieved given the standard deviation of acoustic measurements.

6.3 Data Logging Standard

Data Stream	Rate	Format	Storage	Retention
Battery V/I (Mauch HALL)	100 Hz	MAVLink BATTERY_STATUS	SD card + GCS log	All flights, 5 years
Motor RPM/current (ESC telemetry)	100 Hz	DSHOT ESC_STATUS	SD card	All flights, 5 years
GPS position/velocity	10 Hz	MAVLink GPS_RAW_INT	SD card + GCS	All flights, 5 years
IMU accelerometer/gyro	400 Hz	MAVLink RAW_IMU	SD card	All flights, 2 years
Companion CPU/GPU load	1 Hz	Custom ROS2 topic	Jetson storage	All Jetson flights, 2 years
Acoustic SPL (flight test)	Continuous	WAV 96kHz at 1m ground	Portable recorder	All noise tests, 10 years
Video (front camera)	30fps	H.264 MP4	Jetson storage	Mission flights, 1 year
Wind speed/direction	1 Hz	CSV from anemometer	GCS PC	All outdoor flights, 5 years
Pre/post-flight meta	Per flight	JSON: temp, humidity, battery cycle, operator ID	Fleet database	Permanent

Table 10. Data logging standard for all validation activities.

7. TEST REPORT TEMPLATE

7.1 Standard Report Fields

Every test, bench or flight, produces a standardised report containing:

Section	Required Content
Header	Report ID (format: [SITE]-[DATE]-[TEST_ID]); drone serial(s); operator names; weather conditions
Test objective	One sentence stating what requirement is being verified and the acceptance criterion
Setup photograph	Photograph or diagram of test setup showing all instruments and drone configuration
Procedure as-run	Any deviations from published procedure; equipment substitutions with calibration dates
Raw data table	All measurement values before processing; include outliers and their disposition
Statistical analysis	Mean $\pm$ SD; confidence interval; ANOVA results if applicable; p-values
Pass/fail decision	Explicit GO / NO-GO for the acceptance criterion with margin noted
Anomaly log	Any unexpected behaviour, even if test passed; root cause if known
Next action	For PASS: confirm requirement closed; For FAIL: root cause analysis, re-test date
Sign-off	Pilot-in-command and safety observer signatures with date

Table 11. Standard test report template fields. All sections mandatory for L4/L5 tests.

